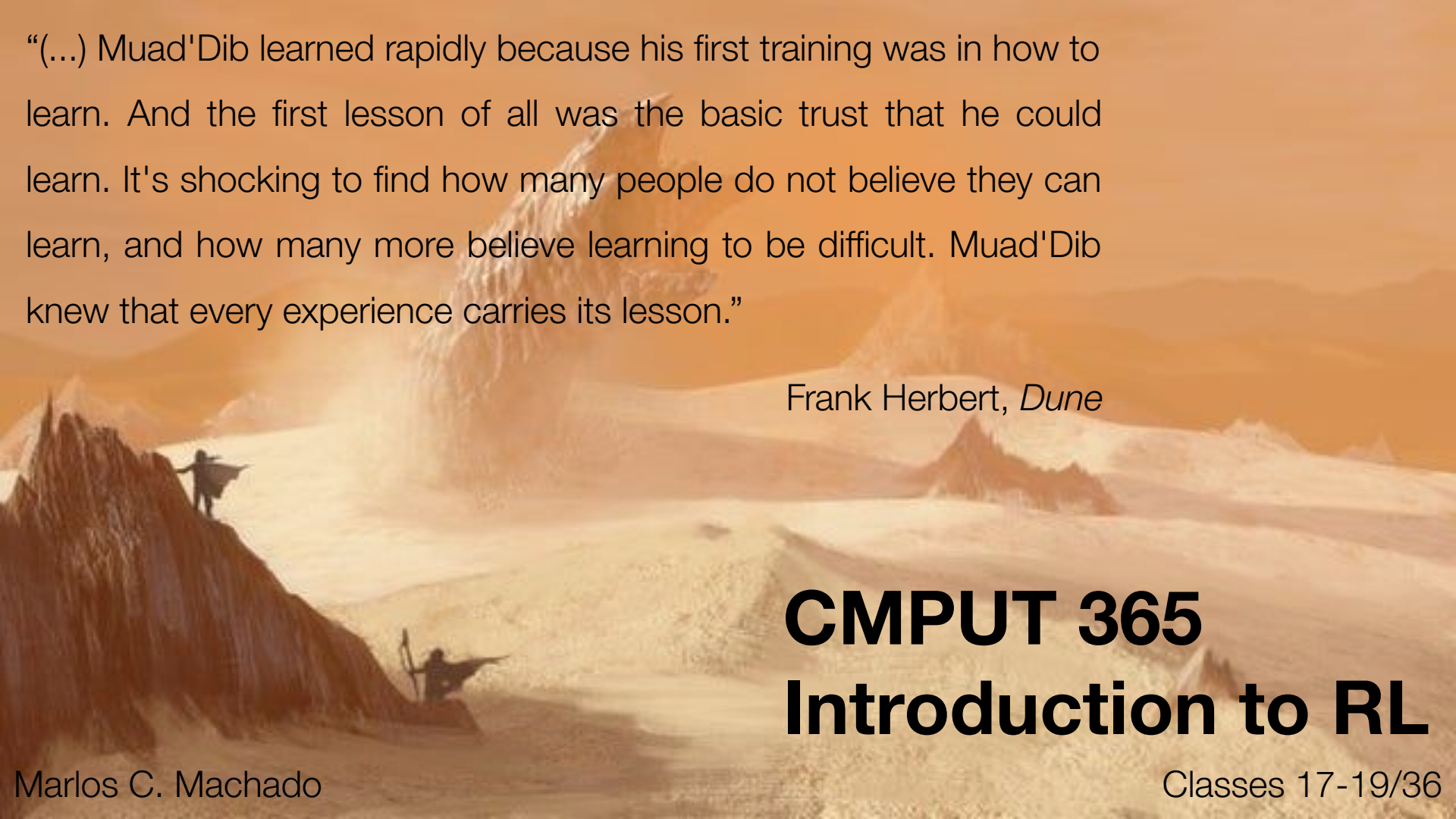




“(...) Muad'Dib learned rapidly because his first training was in how to learn. And the first lesson of all was the basic trust that he could learn. It's shocking to find how many people do not believe they can learn, and how many more believe learning to be difficult. Muad'Dib knew that every experience carries its lesson.”

Frank Herbert, *Dune*

CMPUT 365

Introduction to RL

Coursera Reminder

You **should be enrolled in the private session** we created in Coursera for CMPUT 365.

I **cannot** use marks from the public repository for your course marks. You **need to check, every time**, if you are in the private session and if you are submitting quizzes and assignments to the private section.

At the end of the term, I **will not port grades** from the public session in Coursera.

If you have any questions or concerns, **talk with the TAs** or email us `cmput365@ualberta.ca`.

Reminders and Notes

- Exam viewing:
 - It happened yesterday, today, and will happen tomorrow (from 2pm to 4pm @UCOMM-3-480)
- What I plan to do today:
 - Discuss mid-term course evaluation
 - Start overview of TD Learning Materials
- Useful information for you:
 - The grades for the Monte Carlo
 - The Quiz and Programming Assignment for Temporal Difference Learning is due on Friday.

Re-evaluation of midterm exams: Students will have access to their midterm exam during an exam viewing period. A student who has concerns about how specific questions of their midterm exam were marked can submit a request to the instructor via email within two weeks of the date they received their marked exam. The request should specify (1) which question is to be re-evaluated, (2) the rationale for such a request, and (3) the proposed marks. Importantly, once a request for re-evaluation is submitted, it is up to the instructor's discretion to adjust the marks. *Students won't be allowed to take their midterm exams with them, nor to take pictures of them, so in case of concerns, the student is advised to take notes during the exam viewing period. The TAs are not authorized to weigh in on the midterm exams, this is something only the instructor can do. Notice marks can also go down once a question is re-evaluated.*

DEMYSTIFYING GRAD SCHOOL



UNIVERSITY OF ALBERTA Department of Computing Science
Equity, Diversity, and Inclusion Committee

Registration form for: Demystifying Grad School Workshop 2025

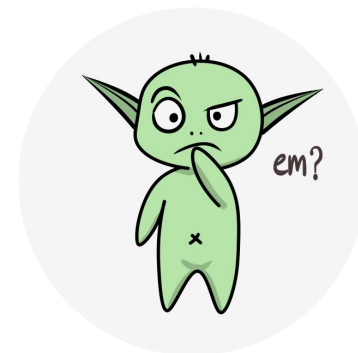
The Department of Computing Science Equity, Diversity, and Inclusion (EDI) committee is once again offering a workshop to help students understand how graduate school works and how students can prepare a strong graduate school application. The workshop will also include a panel of current graduate students and a second panel of faculty members.

When: Tuesday, October 28th, 2025, 5-8pm

Where: Hybrid - please see below

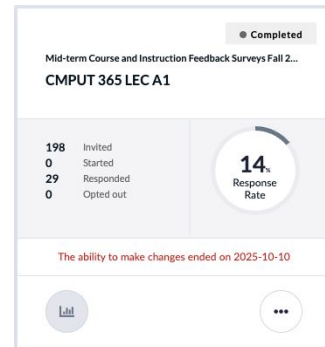
- In-person: University Commons 1-330 (pizza for registered attendees served at 5pm)
- Zoom: <https://ualberta-ca.zoom.us/j/99342601133?pwd=mauUVbO2MZJ8oWU9LbYn8DQToqFMDt.1>

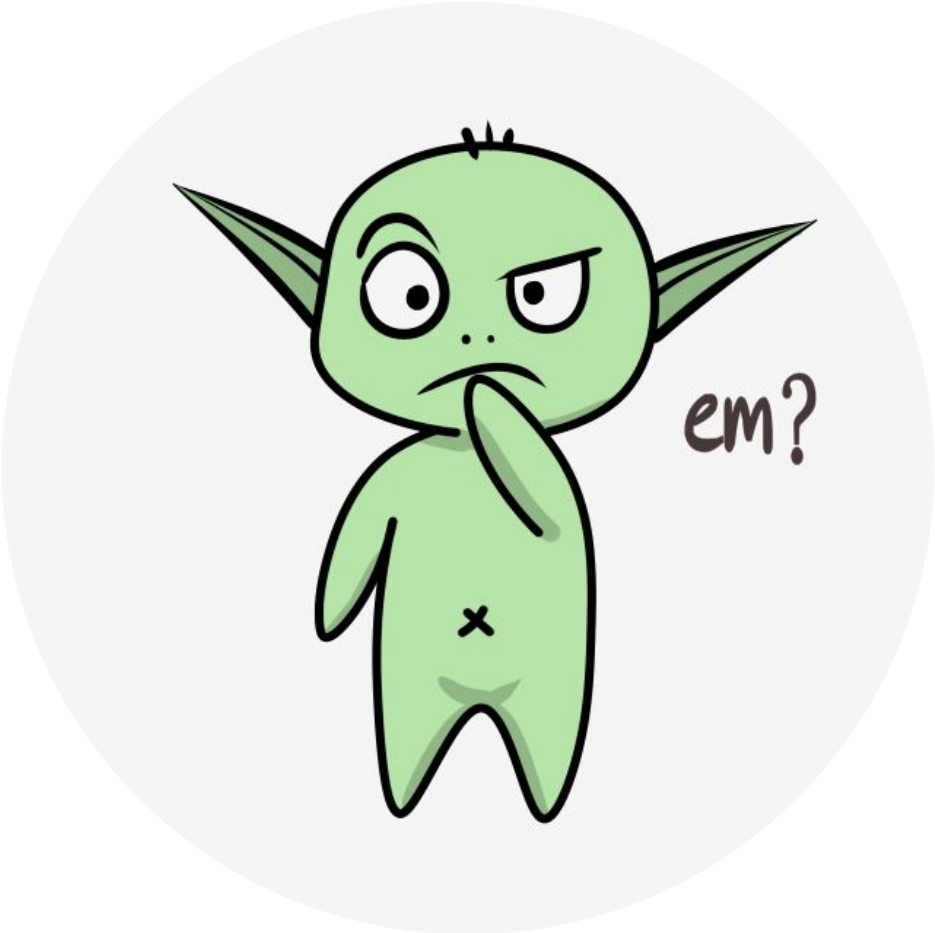
Please, interrupt me at any time!



The deadline for the Mid-term Course Evaluation is passed

- Thank you to those who answered it.
- I want to discuss the responses since you took the time to give the feedback.
- *Overall it was good. That being said, it is quite hard to please everyone, and some are upset about the midterm.*





On studying

- Apparently many people think the course is difficult.
 - I (and the TAs) have been privately approached and asked for advice on how to better study.
- Here's my suggested list of weekly tasks for you:
 1. Before you come to the first class on the topic:
 - a. Read the textbook.
 - b. Watch the Coursera videos and do the Quiz and Programming Assignments.
 2. Come to every class and, during that “week”:
 - a. Do all the exercises of the corresponding Worksheet.

Chapter 6

Temporal-Difference Learning

Prediction

Temporal-difference learning – Why?

“If one had to identify one idea as central and novel to reinforcement learning, it would undoubtedly be temporal-difference (TD) learning.”

TD Prediction

A simple every-visit Monte Carlo method is:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[\underline{G_t} - V(S_t) \right]$$

What if we don't want to wait until we have a full return (end of episode)!

$$NewEstimate \leftarrow OldEstimate + StepSize \left[Target - OldEstimate \right]$$

What can we use instead of G_{t+1} ?

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_t = R_{t+1} + \gamma G_{t+1}$$

$$v_{\pi}(s) = \mathbb{E}_{\pi} [G_t \mid S_t = s]$$

$$v_{\pi}(s) = \mathbb{E}_{\pi} [R_{t+1} + \gamma G_{t+1} \mid S_t = s]$$

$$v_{\pi}(s) = \mathbb{E}_{\pi} [R_{t+1} + \gamma v_{\pi}(S_{t+1}) \mid S_t = s]$$

TD Prediction

A simple every-visit Monte Carlo method is:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[\underbrace{G_t}_{\text{Target}} - V(S_t) \right]$$

Temporal-Difference Learning:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[\underbrace{R_{t+1} + \gamma V(S_{t+1})}_{\text{Target}} - V(S_t) \right]$$

TD Prediction

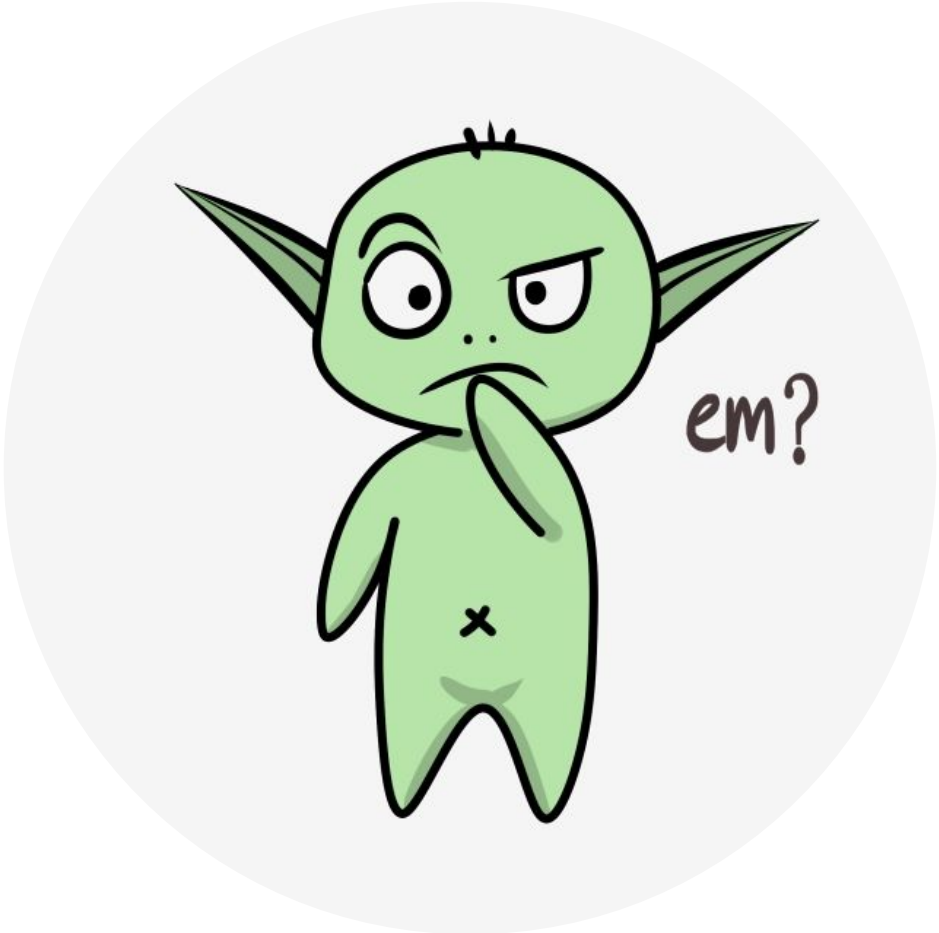
A simple every-visit Monte Carlo method is:

$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)]$$

Temporal-Difference Learning (specifically, **one-step TD**, or **TD(0)**):

$$V(S_t) \leftarrow V(S_t) + \alpha [R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

These are estimates all the way down...



Tabular TD(0)

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated

Algorithm parameter: step size $\alpha \in (0, 1]$

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop for each episode:

 Initialize S

 Loop for each step of episode:

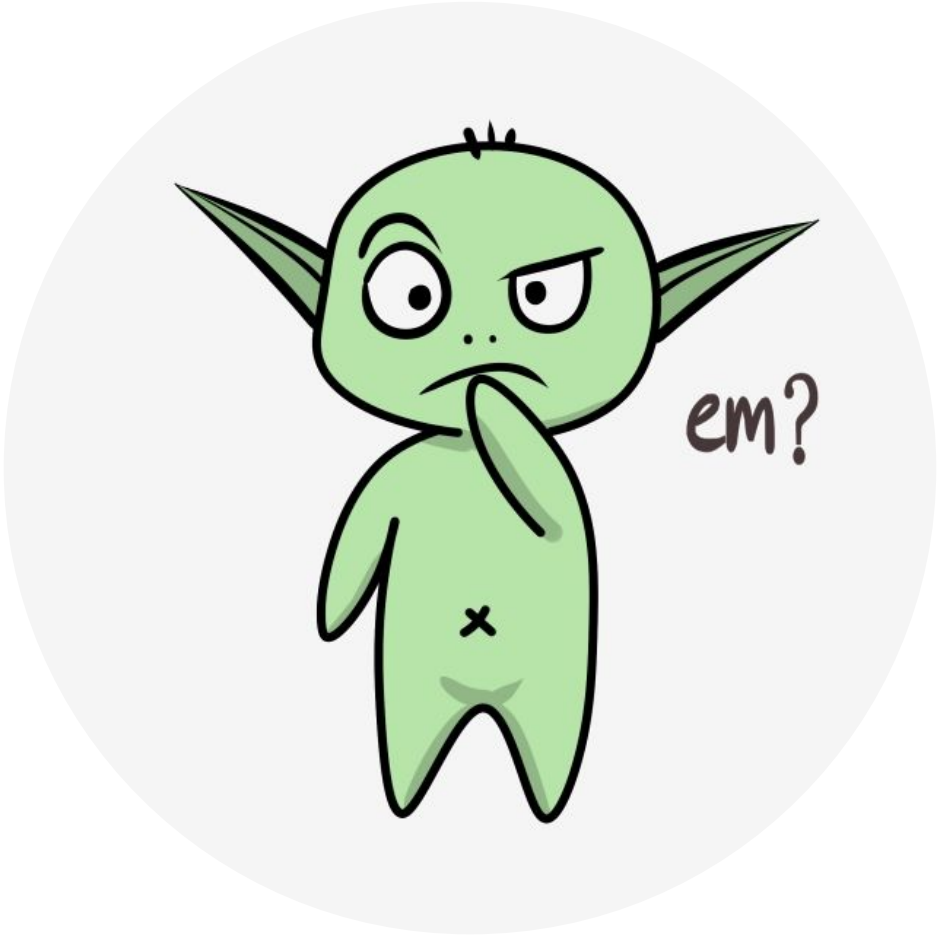
$A \leftarrow$ action given by π for S

 Take action A , observe R, S'

$V(S) \leftarrow V(S) + \alpha [R + \gamma V(S') - V(S)]$

$S \leftarrow S'$

**Sample
update** until S is terminal



Temporal-Difference Error

$$\delta_t \doteq R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

Temporal-Difference Error

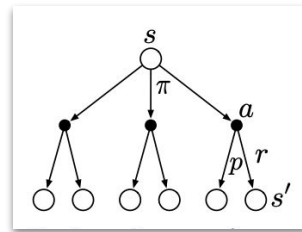
$$V(S_t) \leftarrow V(S_t) + \alpha \left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \right]$$

$$\delta_t \doteq R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

TD is a sample update with bootstrapping

- Dynamic programming update:

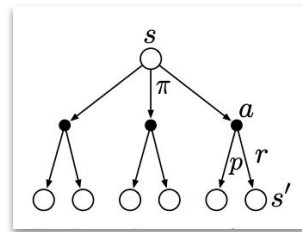
$$\begin{aligned} v_{k+1}(s) &\doteq \mathbb{E}_{\pi}[R_{t+1} + \gamma v_k(S_{t+1}) \mid S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_k(s')] \end{aligned}$$



TD is a sample update with bootstrapping

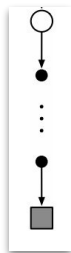
- Dynamic programming update:

$$\begin{aligned} v_{k+1}(s) &\doteq \mathbb{E}_{\pi}[R_{t+1} + \gamma v_k(S_{t+1}) \mid S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_k(s')] \end{aligned}$$



- Monte Carlo update:

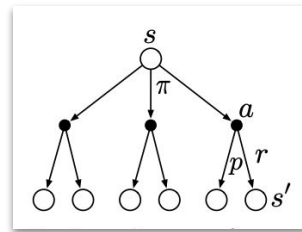
$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)]$$



TD is a sample update with bootstrapping

- Dynamic programming update:

$$\begin{aligned} v_{k+1}(s) &\doteq \mathbb{E}_{\pi}[R_{t+1} + \gamma v_k(S_{t+1}) \mid S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_k(s')] \end{aligned}$$



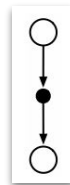
- Monte Carlo update:

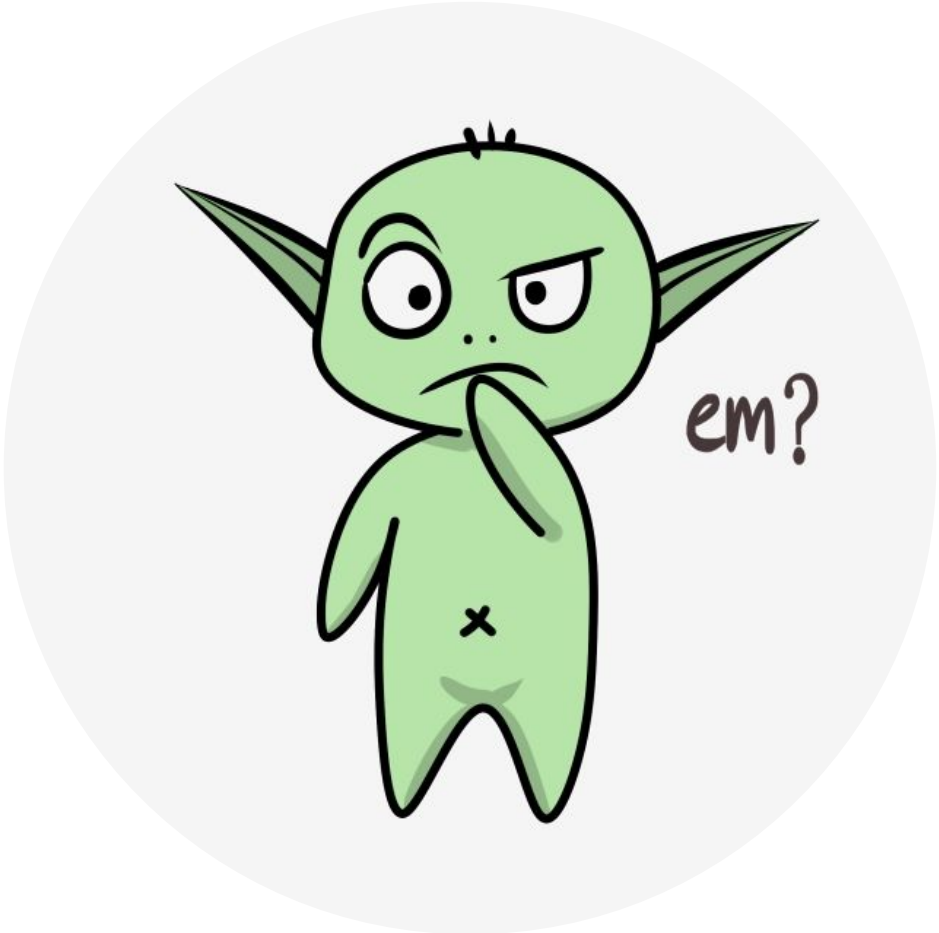
$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)]$$



- TD update:

$$V(S_t) \leftarrow V(S_t) + \alpha [R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$





Example – Driving Home

Example 6.1: Driving Home Each day as you drive home from work, you try to predict how long it will take to get home. When you leave your office, you note the time, the day of week, the weather, and anything else that might be relevant. Say on this Friday you are leaving at exactly 6 o'clock, and you estimate that it will take 30 minutes to get home. As you reach your car it is 6:05, and you notice it is starting to rain. Traffic is often slower in the rain, so you reestimate that it will take 35 minutes from then, or a total of 40 minutes. Fifteen minutes later you have completed the highway portion of your journey in good time. As you exit onto a secondary road you cut your estimate of total travel time to 35 minutes. Unfortunately, at this point you get stuck behind a slow truck, and the road is too narrow to pass. You end up having to follow the truck until you turn onto the side street where you live at 6:40. Three minutes later you are home.

Example – Driving Home

The sequence of states, times, and predictions is thus as follows:

<i>State</i>	<i>Elapsed Time (minutes)</i>	<i>Predicted Time to Go</i>	<i>Predicted Total Time</i>
leaving office, friday at 6	0	30	30
reach car, raining	5	35	40
exiting highway	20	15	35
2ndary road, behind truck	30	10	40
entering home street	40	3	43
arrive home	43	0	43

Example – Driving Home

The rewards in this example are the elapsed times on each leg of the journey.¹ We are not discounting ($\gamma = 1$), and thus the return for each state is the actual time to go from that state. The value of each state is the *expected* time to go. The second column of numbers gives the current estimated value for each state encountered.

The sequence of states, times, and predictions is thus as follows:

<i>State</i>	<i>Elapsed Time (minutes)</i>	<i>Predicted Time to Go</i>	<i>Predicted Total Time</i>
leaving office, friday at 6	0	30	30
reach car, raining	5	35	40
exiting highway	20	15	35
2ndary road, behind truck	30	10	40
entering home street	40	3	43
arrive home	43	0	43

Example – Driving Home

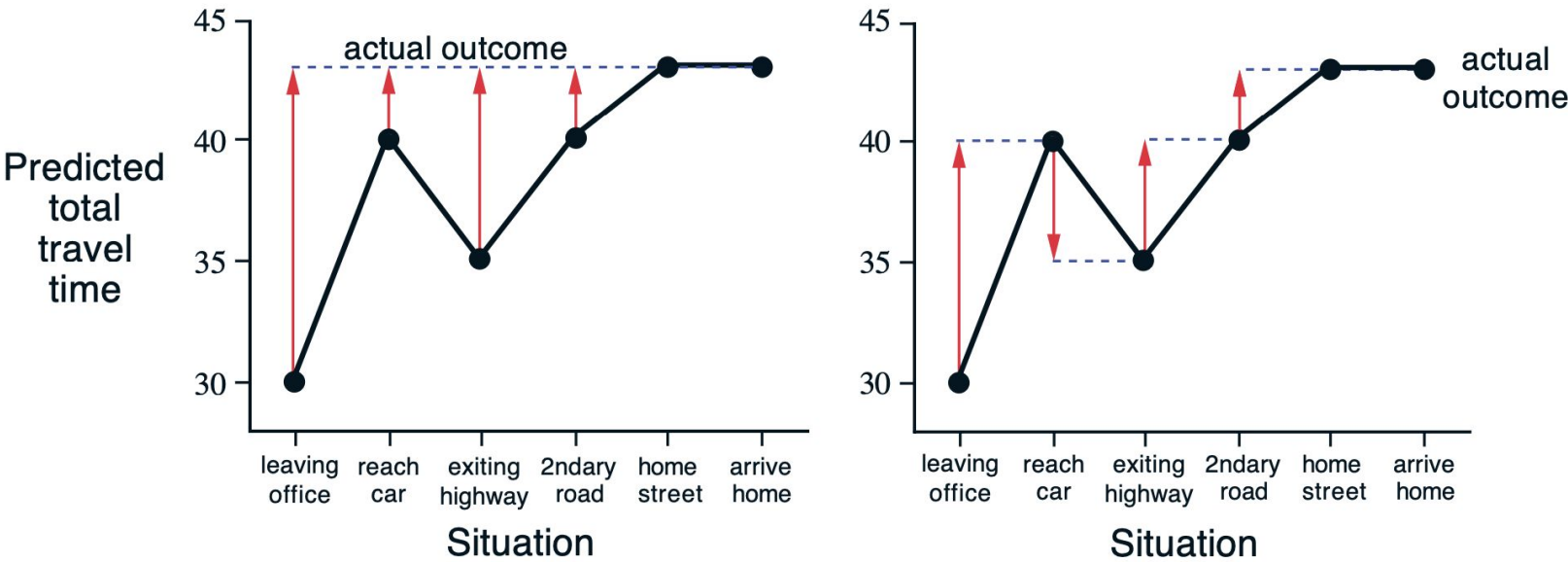
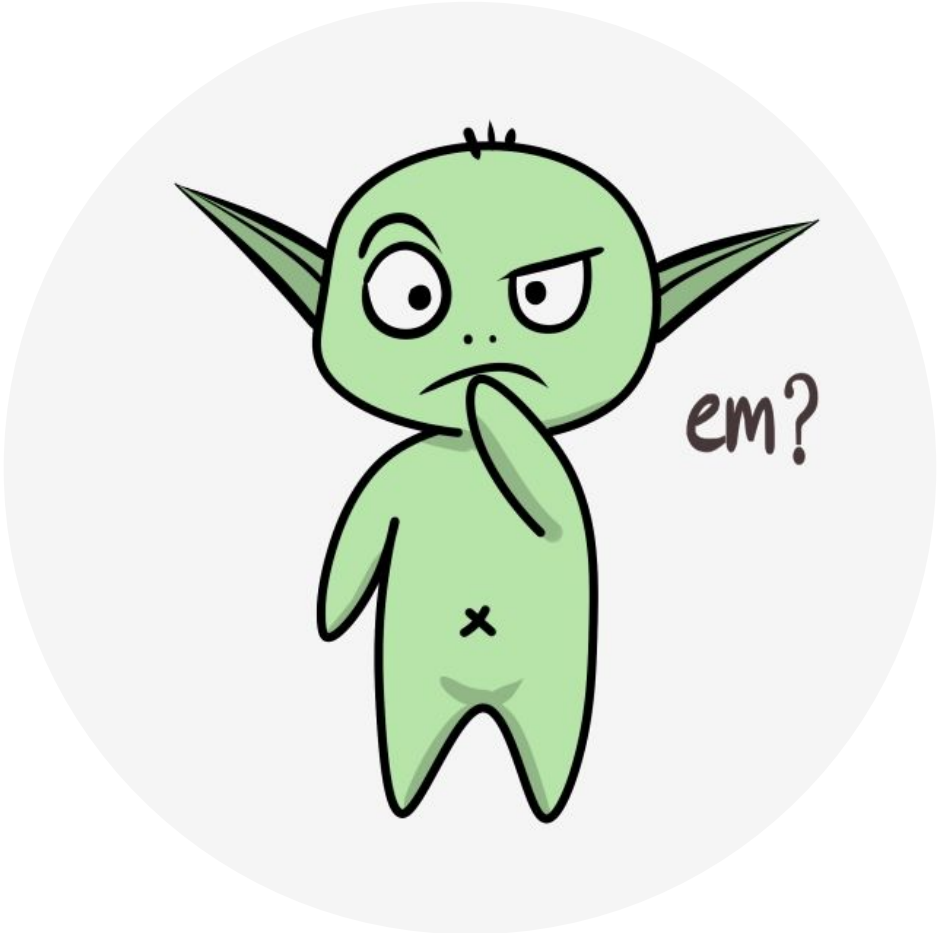


Figure 6.1: Changes recommended in the driving home example by Monte Carlo methods (left) and TD methods (right).



Optimality of TD(0)

Example 6.4: You are the Predictor Place yourself now in the role of the predictor of returns for an unknown Markov reward process. Suppose you observe the following eight episodes:

A, 0, B, 0

B, 1

B, 1

B, 1

B, 1

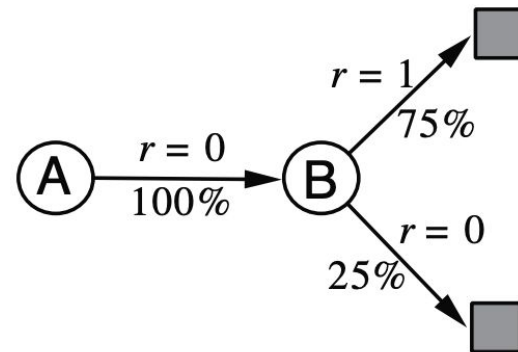
B, 1

B, 1

B, 0

$V(A) = ?$

$V(B) = ?$



Optimality of TD(0)

Example 6.4: You are the Predictor Place yourself now in the role of the predictor of returns for an unknown Markov reward process. Suppose you observe the following eight episodes:

A, 0, B, 0

B, 1

B, 1

B, 1

B, 1

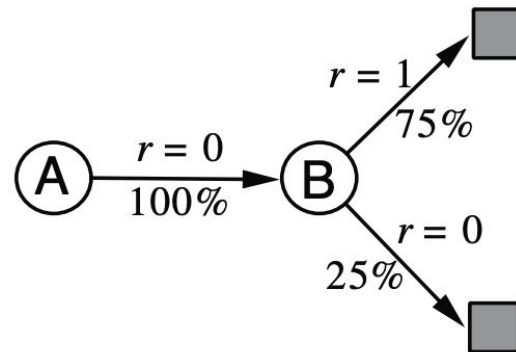
B, 1

B, 1

B, 0

$V(A) = ?$ **TD** **MC**
 $\frac{3}{4}$ or 0?

$V(B) = \frac{3}{4}$



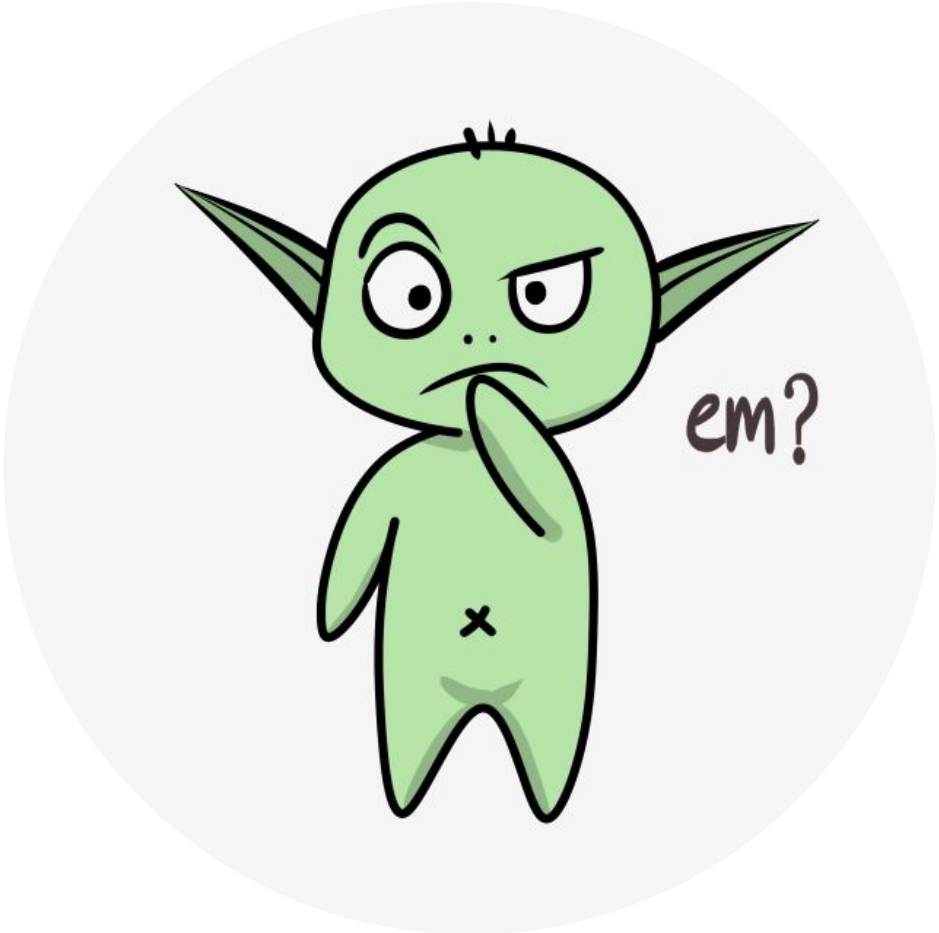
Optimality of TD(0)

- Under batch training, constant- α MC converges to values, $V(s)$, that are sample averages of the actual returns experienced after visiting each state s . These are optimal estimates in the sense that they minimize the mean square error from the actual returns in the training set.
- But TD(0) gives us the answer that it is based on first modeling the Markov process and then computing the correct estimates given the model (the *certainty-equivalence estimate*).

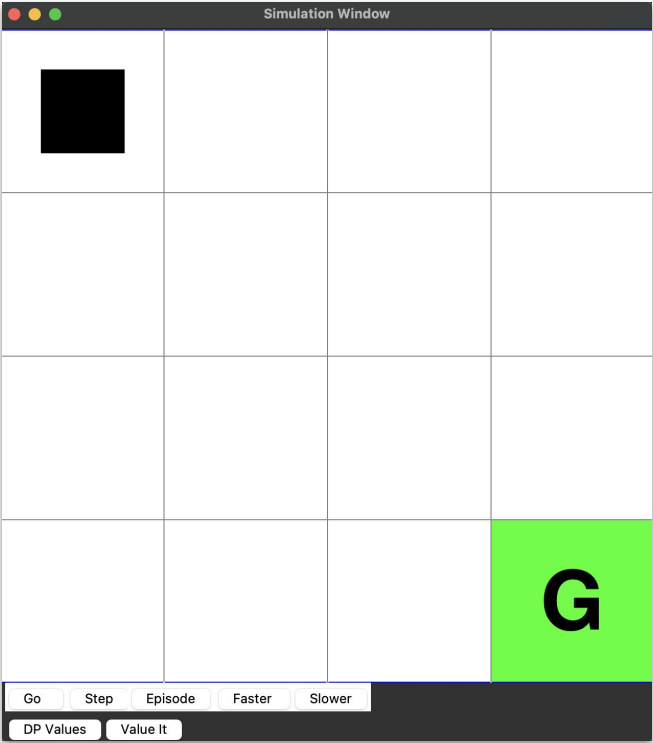
TD vs Monte Carlo

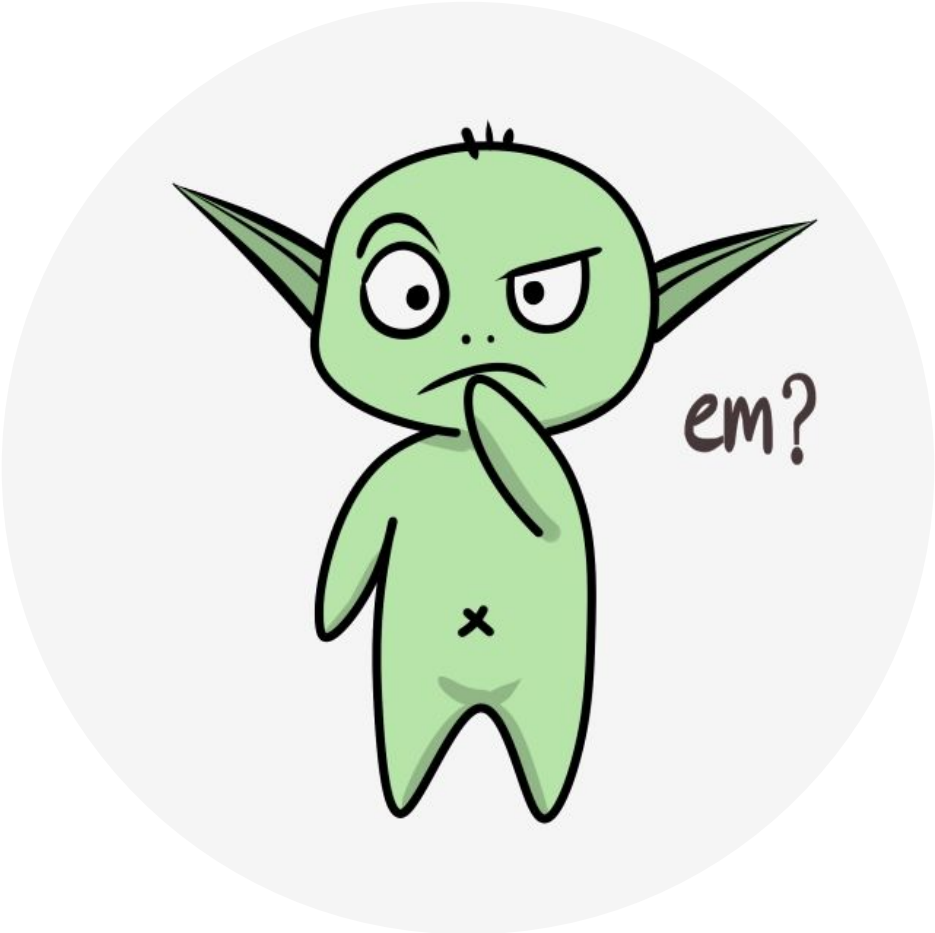
“Batch Monte Carlo methods always find the estimates that minimize mean square error on the training set, whereas batch TD(0) always finds the estimates that would be exactly correct for the maximum-likelihood model of the Markov process.”

In general, the *maximum-likelihood estimate* of a parameter is the parameter value whose probability of generating the data is greatest.



Demo





Reward Prediction Error Hypothesis in Neuroscience

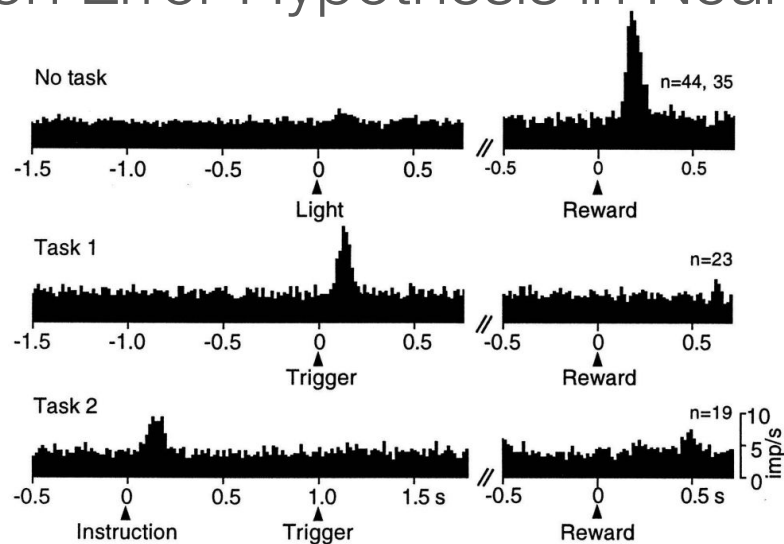
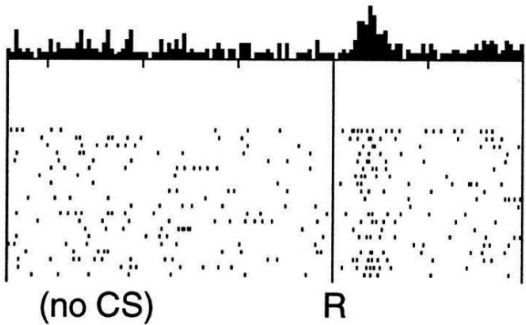


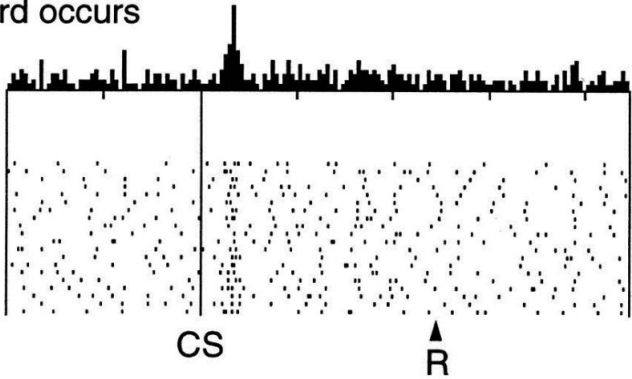
Figure 15.2: The response of dopamine neurons shifts from initial responses to primary reward to earlier predictive stimuli. These are plots of the number of action potentials produced by monitored dopamine neurons within small time intervals, averaged over all the monitored dopamine neurons (ranging from 23 to 44 neurons for these data). Top: dopamine neurons are activated by the unpredicted delivery of drop of apple juice. Middle: with learning, dopamine neurons developed responses to the reward-predicting trigger cue and lost responsiveness to the delivery of reward. Bottom: with the addition of an instruction cue preceding the trigger cue by 1 second, dopamine neurons shifted their responses from the trigger cue to the earlier instruction cue. From Schultz et al. (1995), MIT Press.

Reward Prediction Error Hypothesis in Neuroscience

No prediction
Reward occurs



Reward predicted
Reward occurs



Reward predicted
No reward occurs

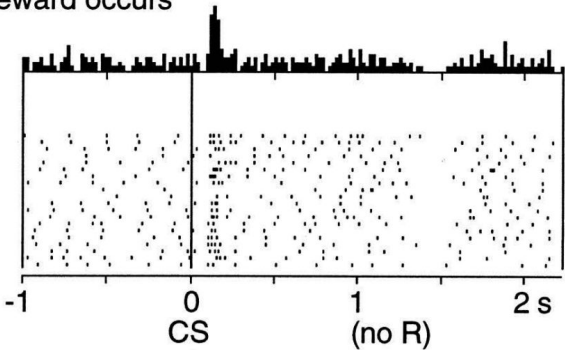
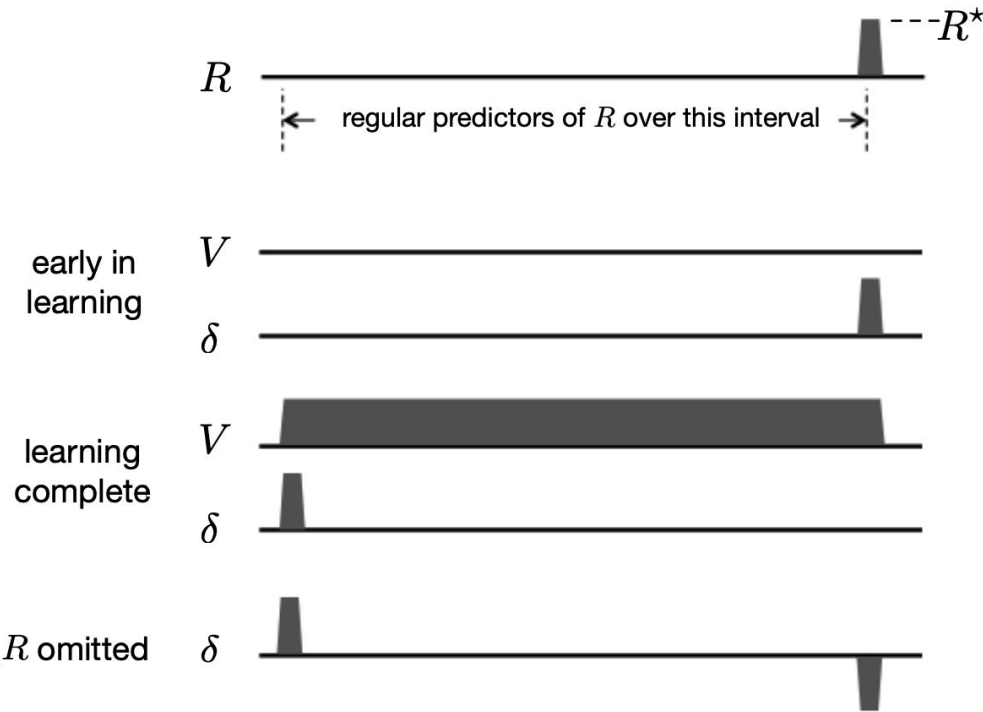
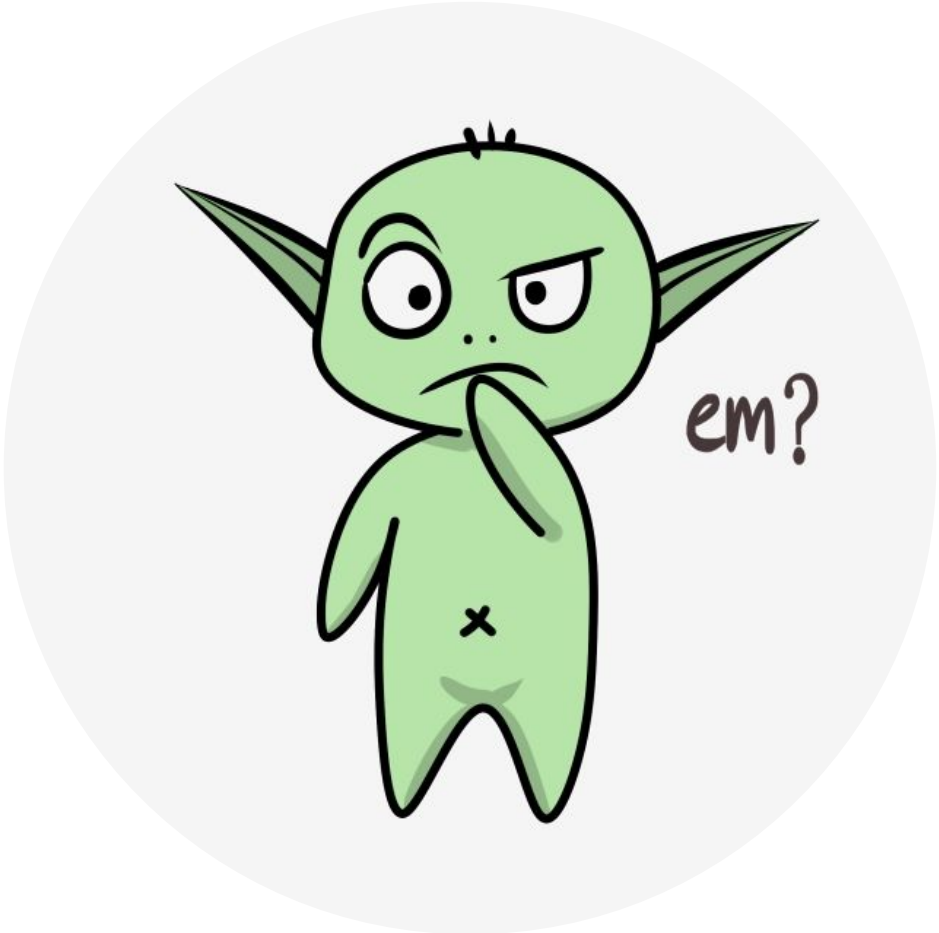


Figure 15.3: The response of dopamine neurons drops below baseline shortly after the time when an expected reward fails to occur. Top: dopamine neurons are activated by the unpredicted delivery of a drop of apple juice. Middle: dopamine neurons respond to a conditioned stimulus (CS) that predicts reward and do not respond to the reward itself. Bottom: when the reward predicted by the CS fails to occur, the activity of dopamine neurons drops below baseline shortly after the time the reward is expected to occur.

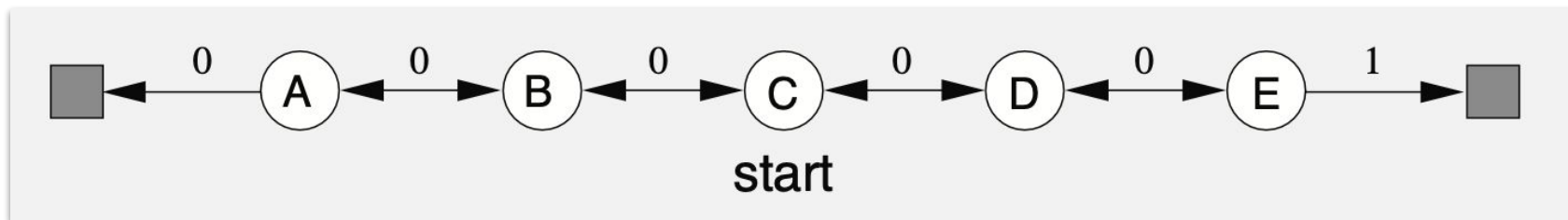
At the top of each of these panels is shown the average number of action potentials produced by monitored dopamine neurons within small time intervals around the indicated times. The raster plots below show the activity patterns of the individual dopamine neurons that were monitored; each dot represents an action potential. From Schultz, Dayan, and Montague, A Neural Substrate of Prediction and Reward, *Science*, vol. 275, issue 5306, pages 1593-1598, March 14, 1997. Reprinted with permission from AAAS.

TD Error/Dopamine Correspondence



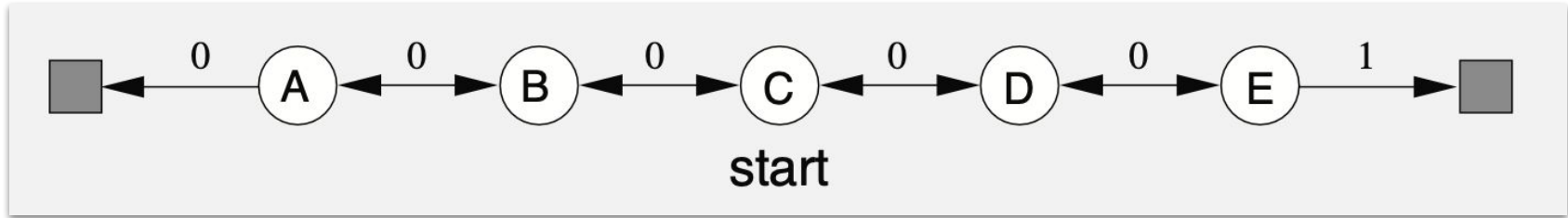


Example / Exercise: Random Walk



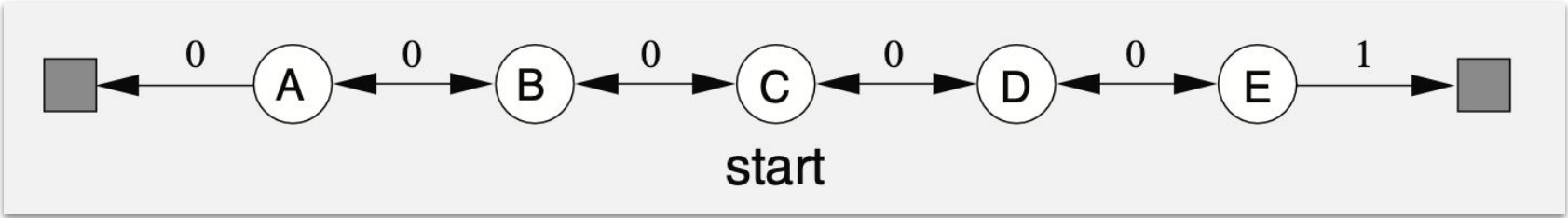
- *Markov reward process* (MRP), an MDP without actions.
- Start state: C
- One can go left or right, on each step, with equal probability.
- Reward: +1 on right exit, 0 otherwise.
- $\gamma = 1.0$
- All values are initialized with 0.5, that is $V(s) = 0.5$ for all s .

Example / Exercise: Random Walk

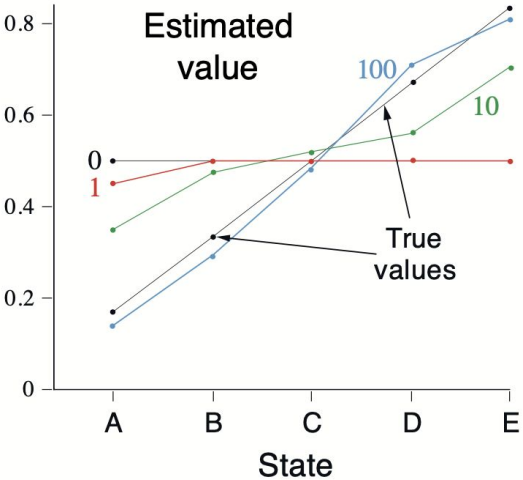


Q1. What does v_{π} encode in this problem?

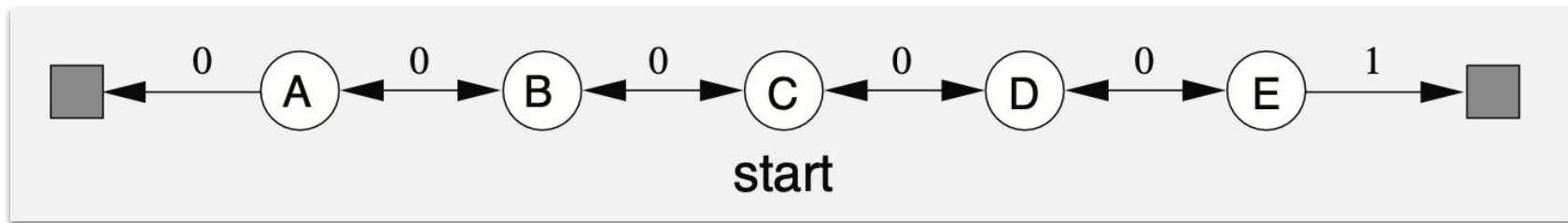
Example / Exercise: Random Walk



Q1. What does v_{π} encode in this problem?



Example / Exercise: Random Walk



Q1. What does v_{π} encode in this problem?

Q2. The first episode results in a change in only $V(A)$. What does this tell you about what happened on the first episode?

Q3. Why was only the estimate for this one state changed?

Q4. By exactly how much was it changed? (Assume $\alpha=0.1$)

